Mesh Radio v1 (working name) — OffGrid Devices
LoRa 902–928 MHz mesh radio, up to ~3 W; Bluetooth LE; one LiPo cell; USB-C, solar and drone power.
NDAA-ready parts list: every part from a US / EU / Japan / Taiwan / Korea / Norway maker.
Each box below is one page. Every connection is a named label; the same name is the same wire.
This drawing is generated from src/circuit.py by src/sch.py: change the code, not the drawing.

Power in: USB-C, solar, drone 5 V (each surge-clamped and fused)

USB: 22 V flat-clamp TVS, a 4.7 uF + 1 R snubber and a TPS259470A eFuse (cut-off 5.4 V, 1.5 A). Solar and drone: 27 V TVS and 60 mA PTC (cut-off 10 V, 0.71 A). Diode-OR'd to VBUS, with reverse-current blocking.

File: inputs.kicad_sch

Charger and battery: protection, reverse polarity, battery sense

Battery: 2-cell (7.4 V) pack, 4.2 A MAX17502R (CCD ~5.8 A, SCD ~29 A, 0.71 mΩ). Ideal-diode OR'd with the VBUS path; charging only when the cell is the lower rail.

File: charger.kicad_sch

System power: on/off switch, 3.3 V rail

3.3 V, 2.5 A TPS62743 buck-boost, always on from the battery. Power-path: VBUS feeds the 3.3 V rail directly when present, so the system runs even with a dead cell.

File: system.kicad_sch

Processor and Bluetooth: nRF52840

nRF52840 in a Raytac MDBT50Q-1MV2 module (+8 dBm, 1 MB flash, 256 kB RAM). Powered from the 3.3 V rail, so its pins and the SX1262 share one logic level.

File: mcu.kicad_sch

LoRa radio: SX1262, its clock, transmit match and receive balun

Semtech SX1262MB2CAS (915 MHz) reference values. TCXO on DIO3 (set 2.2 V in firmware), DIO2 drives the antenna switch.

File: radio.kicad_sch

RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA

Driver: SX1262 +11 dBm -> 30 dBm (1 W), +17 dBm -> ~35 dBm (3 W). The 15.9 dB pad keeps the amplifier input matched.

File: frontend.kicad_sch

Amplifier power: battery-first supply, buck-boost, control voltage, temperature

Amplifier runs from the battery through a buck-boost, so SX1262 max output (+22 dBm) holds down to 2.8 V. Firmware selects charger SYS when

File: pa_power.kicad_sch

Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, buttons, LEDs

Drone port: 5 V, 1 W, 4 A fuse. GPS: 3.3 V, 0.2 W, 1 A fuse. Header: 1 5V, 2 3V3, 3 TX, 4 RX, 5 RTS, 6 GND. Series resistors on

File: ports.kicad_sch

Generated from src/circuit.py by src/sch.py. Edit the circuit, not this file.
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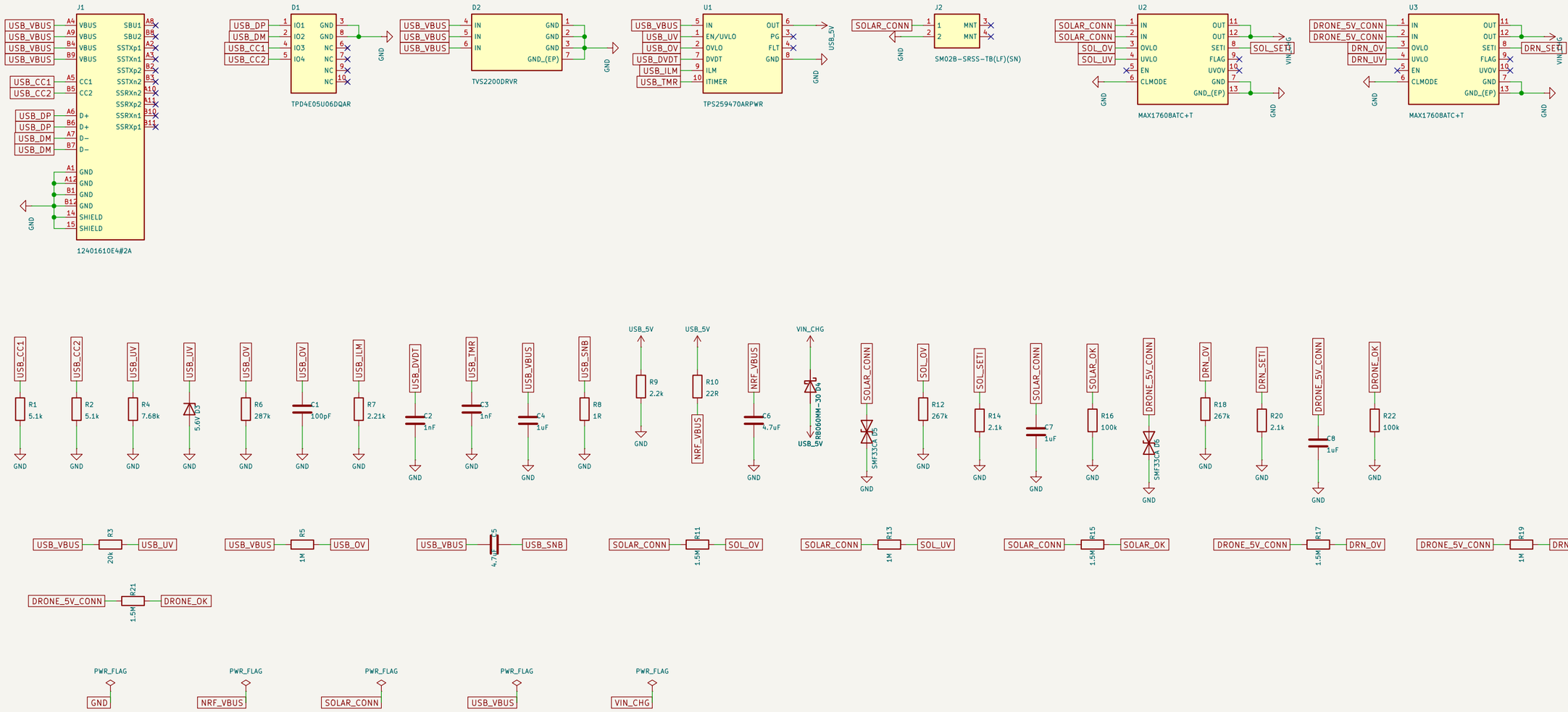
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File: mesh-radio.kicad_sch

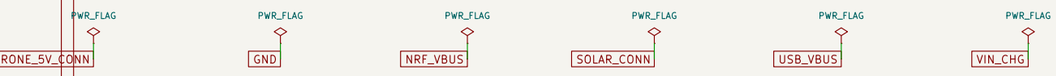
Title: Mesh Radio v1 (working name)

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KiCad E.D.A. 10.0.7		Id: 1/9

Power in: USB–C, solar, drone 5 V (each surge–clamped and fused)

USB: 22 V flat–clamp TVS, a 4.7 uF + 1 R snubber and a TPS259470A eFuse (cut–off 5.4 V, 1.5 A).
Solar and drone: 33 V TVS and a 60 V MAX17608 (cut–off 10 V, 0.71 A, blocks –60 V and reverse current).





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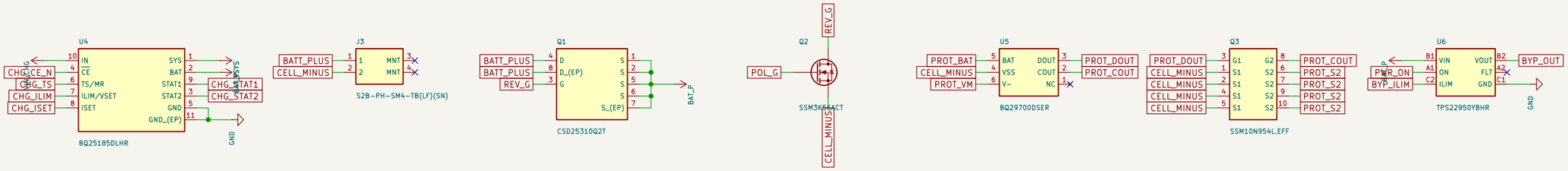
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File: inputs.kicad_sch

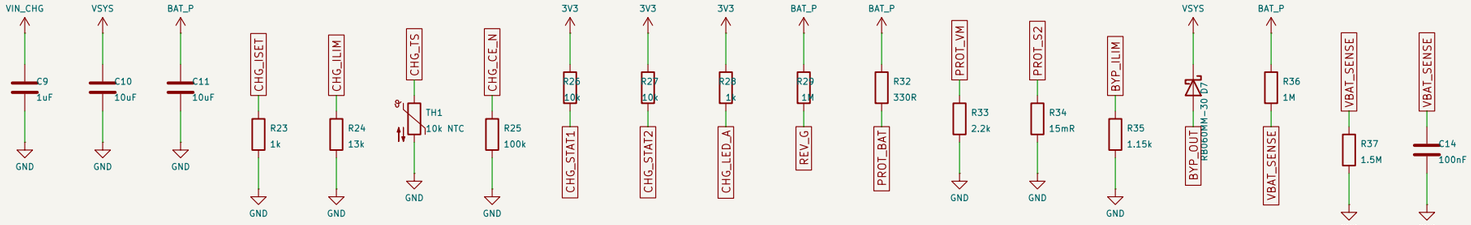
Title: Power in: USB–C, solar, drone 5 V (each surge–clamped and fused)

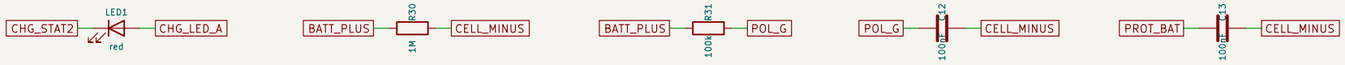
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Charger and battery: protection, reverse polarity, battery sense

Battery plug wired the Adafruit/SparkFun way (pad 1 = +). Cell protection on the minus side (OCD ~5.8 A, SCD ~29 A, OV 4.275 V, UV 2.8 V); reverse-battery P-FET on the plus side, on only when the cell is the right way round. Charge 300 mA. Current-limited cell-to-SYS bypass.









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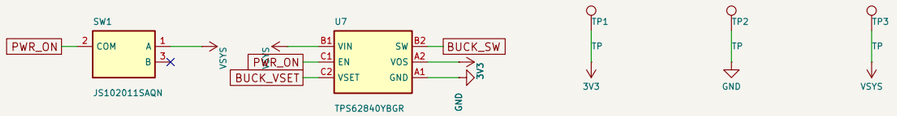
Sheet: /Charger and battery: protection, reverse polarity, battery sense/
File: charger.kicad_sch

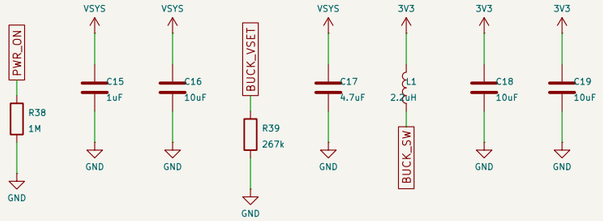
Title: Charger and battery: protection, reverse polarity, battery sense

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System power: on/off switch, 3.3 V rail

The slide switch drives only enables (3.3 V buck, cell bypass); charging works with the board off.





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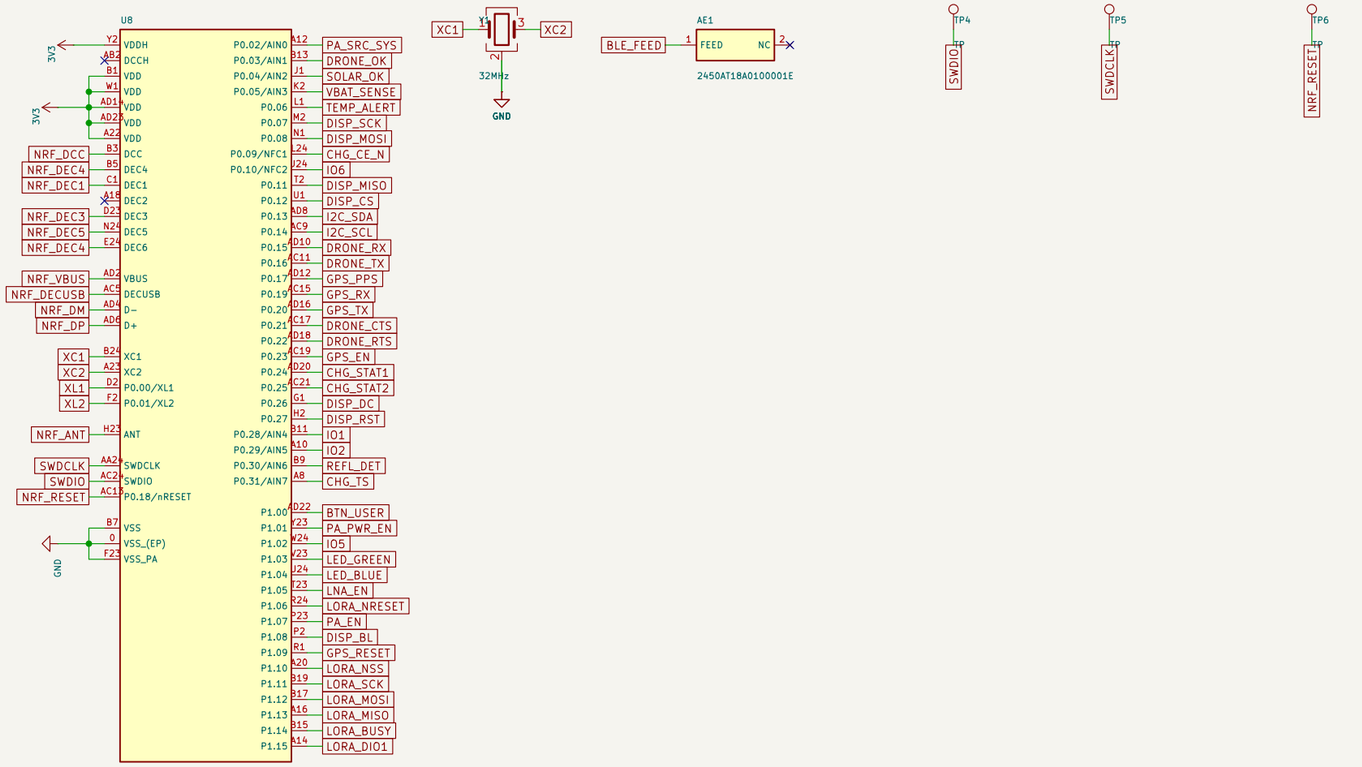
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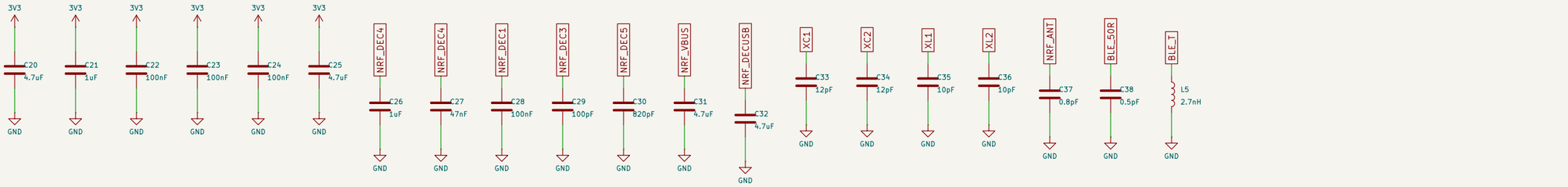
Title: System power: on/off switch, 3.3 V rail

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KiCad E.D.A. 10.0.7		Id: 4/9

Processor and Bluetooth: nRF52840

nRF52840 in normal-voltage mode: VDD = VDDH = the 3.3 V rail, so its pins and the SX1262 share one logic level. REG1 DC/DC on (10 uH + 15 nH). Nordic reference circuit, aQFN73.







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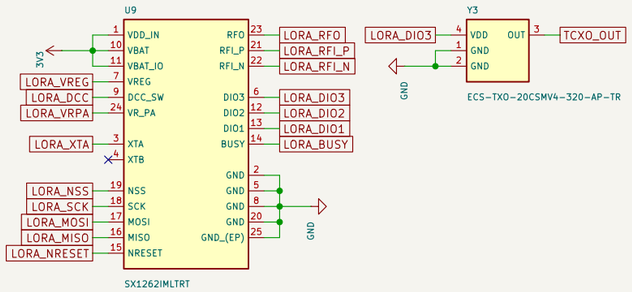
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File: mcu.kicad_sch

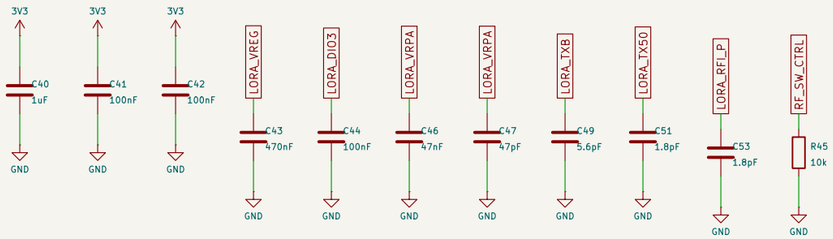
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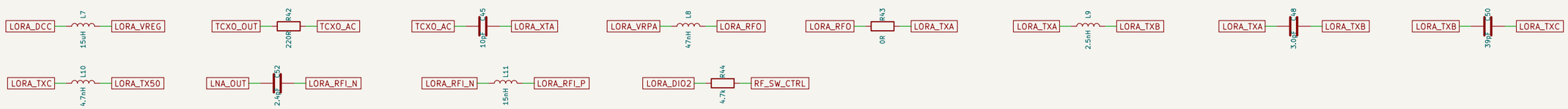
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KiCad E.D.A. 10.0.7		Id: 5/9

LoRa radio: SX1262, its clock, transmit match and receive balun

Semtech SX1262MB2CAS (915 MHz) reference values. TCXO on DIO3 (set 2.2 V in firmware), DIO2 drives the antenna switch.







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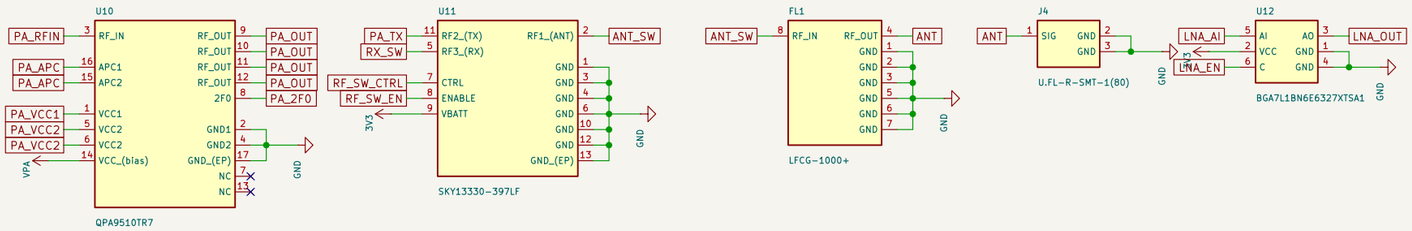
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File: radio.kicad_sch

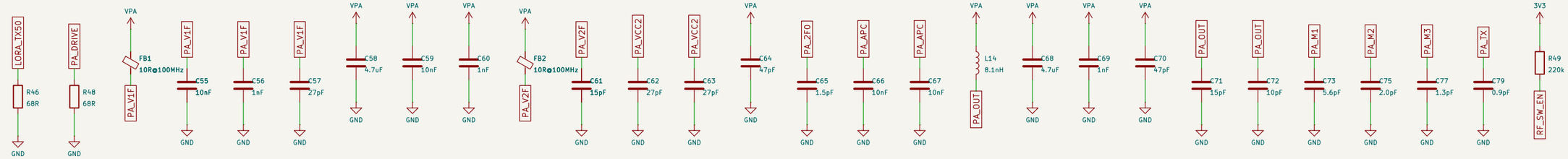
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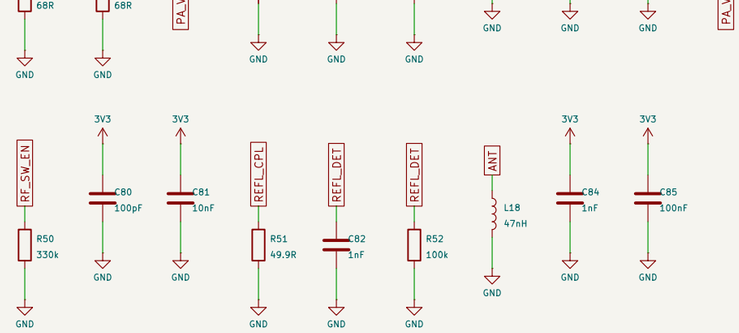
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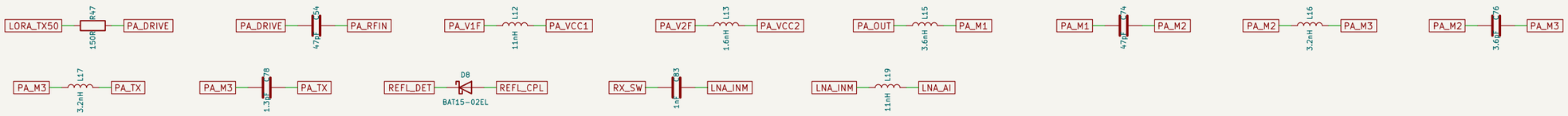
RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA

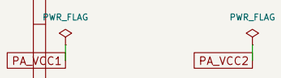
Drive: SX1262 +11 dBm -> 30 dBm (1 W), +17 dBm -> ~35 dBm (3 W). The 15.9 dB pad keeps the amplifier input below its +13 dBm limit even at the SX1262 maximum (+22 dBm -> +6 dBm).











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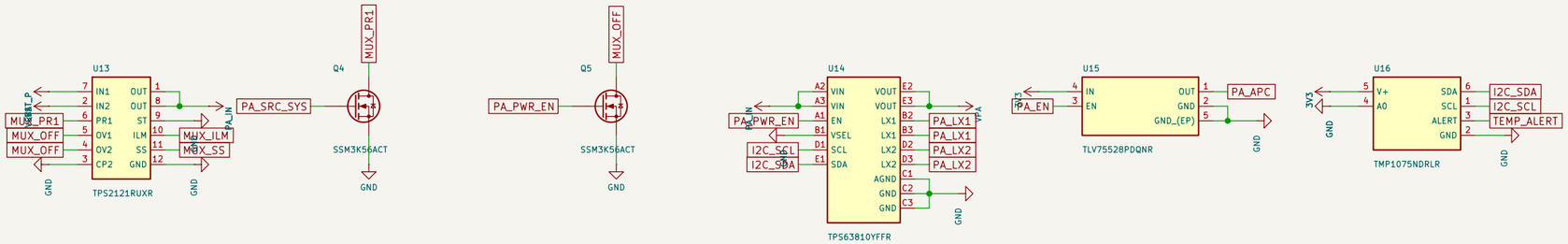
Sheet: /RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA/
File: frontend.kicad_sch

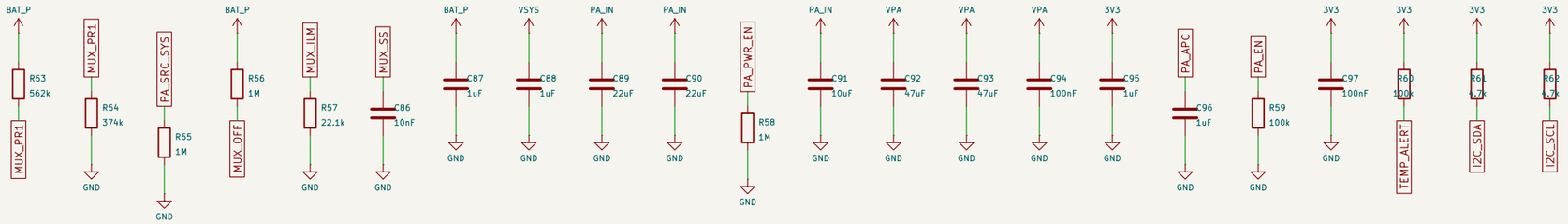
Title: RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA

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Amplifier power: battery–first supply, buck–boost, control voltage, temperature

Amplifier runs from the battery through a power mux (battery first down to 2.6 V; firmware selects
charger SYS when no cell is fitted), then a TPS63810 buck–boost (I2C 0x75): 2.7 V for 1 W, 4.0 V for 3 W.







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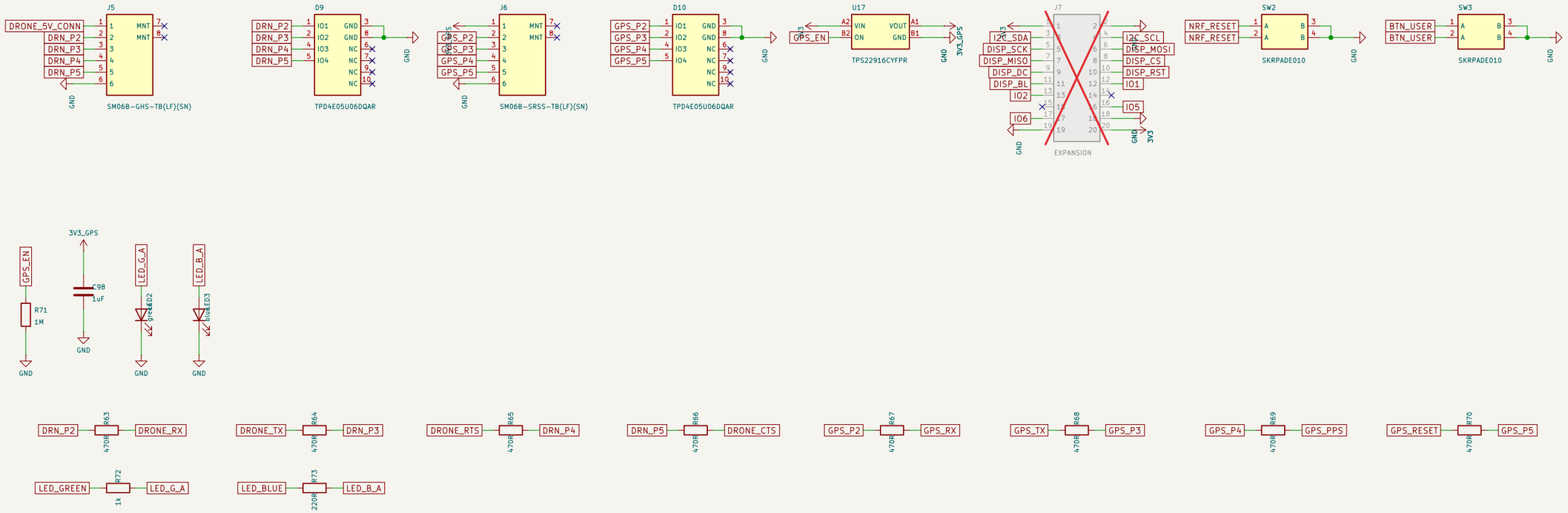
Sheet: /Amplifier power: battery–first supply, buck–boost, control voltage, temperature/
File: pa_power.kicad_sch

Title: Amplifier power: battery–first supply, buck–boost, control voltage, tem

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Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, buttons, LEDs

Drone port wired as Pixhawk TELEM (pins from the autopilot side): 1 5V, 2 TX, 3 RX, 4 CTS, 5 RTS,
6 GND. Series resistors and ESD on every external signal.



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Sheet: /Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, buttons, LEDs/
File: ports.kicad_sch

Title: Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, butto

Size: A3	Date:	**Rev: v1**
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